Short Term Interest Rate Innovations Across Monetary Regimes

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Draft

Status: Preliminary research design. This draft records the research question, replication design, and planned extensions. It does not report empirical estimates, risk prices, standard errors, or pricing-error magnitudes.

Abstract

This paper studies whether short-term interest-rate innovations help explain the cross section of stock-market anomaly returns, and whether that relation is stable across later monetary regimes. The design begins with a strict replication of Maio and Santa-Clara (2017) and separates exact replication, documented reconstruction, and new extension evidence. The planned extensions test temporal stability, regime dependence, weak-factor sensitivity, and the distinction between aggregate rate innovations and identified high-frequency components. The current draft is preliminary: the publication targets and source-access requirements are registered, but source-compatible empirical panels and generated result tables are not yet available. Numerical findings are therefore withheld until the baseline and extension artifacts can be generated and audited.

1 Introduction

Short-term interest rates are plausible state variables in intertemporal asset-pricing models because they summarize investment opportunities that matter for both discount rates and hedging demands. If portfolios formed on accounting or return-based anomalies have different exposures to those state variables, a rate innovation may help organize average excess returns beyond the market factor. The economic question is not only whether such a factor prices anomaly returns in one sample, but whether the relation survives changes in monetary regimes and source definitions.

Maio and Santa-Clara (2017) provide the starting point. Their article studies short-term interest-rate innovations and stock-market anomalies, and this paper uses that design as the baseline object to be reproduced before any extension is interpreted. The strict baseline window is January 1972 through December 2013. The extension window begins in January 2014 and is evaluated separately from the replication verdict.

The paper has three aims. First, it rebuilds the original baseline in a way that distinguishes exact replication from documented reconstruction. Second, it asks whether the short-rate factor is stable after the publication sample and across monetary regimes. Third, it asks whether an aggregate autoregressive rate innovation is sufficient once high-frequency policy and information components are available. The last aim is interpretive: outside an identification design, the baseline residual is a rate innovation, not an identified policy object.

The current evidence status is intentionally limited. The article and supplement targets have been registered, and the software pipeline contains data, estimation, diagnostic, manuscript, and release gates. The empirical inputs needed to populate the baseline tables are still incomplete, so the results sections below state pending-result conditions rather than estimates.

2 Related literature

The first literature stream motivates the state variable. The ICAPM shows why assets can earn premia for covariance with variables that forecast future investment opportunities (Merton, 1973). Interest-rate applications of this idea are directly relevant because Lioui and Maio (2014) model interest-rate risk as a priced source of cross-sectional equity returns, and Maio and Santa-Clara (2017) apply short-rate innovations to anomaly portfolios.

The second stream concerns anomaly pricing and model comparison. The baseline article compares a short-rate ICAPM with standard factor models, so the extension must be evaluated against asset-pricing benchmarks rather than against the CAPM alone. Fama and French (2015) add profitability and investment factors to the Fama-French framework, while Hou et al. (2015) propose an investment approach that summarizes many anomalies with market, size, investment, and profitability factors. These benchmarks make the extension question sharper: does the short-rate factor add robust pricing content once comparator models, asset intersections, and portfolio definitions are held fixed?

The third stream concerns inference. Two-pass cross-sectional tests are exposed to generated-regressor and weak-factor problems. Fama and MacBeth (1973) provide the classic cross-sectional testing framework, Shanken (1992) derives corrections for beta-estimation error, and Kan and Zhang (1999) show that useless factors can appear priced in two-pass tests. Lewellen et al. (2010) argue that high cross-sectional fit and small pricing errors can be misleading when test assets share a strong characteristic structure. This paper therefore treats beta dispersion, rank, spanning, and influence diagnostics as conditions for interpretation rather than optional robustness checks.

The fourth stream concerns monetary regimes and high-frequency identification. Bernanke and Kuttner (2005) measure equity-market responses to unanticipated Federal Reserve policy actions, while Jarociński and Karadi (2020) separate policy and central-bank-information components using high-frequency announcement comovement. Swanson and Williams (2014) show why lower-bound episodes can change the informativeness of short-term rates for the broader term structure. Those papers motivate separating the aggregate AR rate innovation from identified announcement components and from regime-specific pricing evidence. Structural-change methods such as Bai and Perron (1998) provide a way to distinguish predeclared regime boundaries from estimated breaks.

The final stream is data vintage discipline. Official factor archives document that published factor and portfolio series can change as databases and construction rules are revised (French, 2026). This paper therefore adds value beyond the original article by freezing a table-level replication target before asking whether the short-rate evidence generalizes over time, across regimes, and after high-frequency decomposition.

3 Original framework and economic mechanism

The original framework is a discrete-time ICAPM application in which the market excess return is paired with an innovation in a short-term interest-rate state variable. The article verifies two state variables: the federal funds rate and the three-month Treasury-bill rate. It motivates both series as proxies for investment opportunities and constructs innovations from first-order autoregressions.

For the federal-funds version, the article reports the state-variable law of motion as

$$FFR_{t+1} = 0.000 + 0.991 FFR_t, \quad R^2 = 0.98,$$

with the innovation defined as the realized rate less the fitted autoregressive component. The article reports the analogous Treasury-bill equation as

$$TB_{t+1} = 0.000 + 0.992TB_t, \quad R^2 = 0.98.$$

For test asset i, excess return $R^e_{i,t}$, market factor $R_{M,t}$, and federal-funds innovation $\widetilde{FFR}_t$, the verified first-pass time-series regression is

$$R^e_{i,t} = \delta_i + \beta_{i,M} R_{M,t} + \beta_{i,FFR}\widetilde{FFR}_t + \varepsilon_{i,t}.$$

The benchmark second-pass regression is an OLS cross-sectional regression without an intercept:

$$\bar{R}^e_i = \hat{\beta}_{i,M}\lambda_M + \hat{\beta}_{i,FFR}\lambda_{FFR} + \alpha_i^{CS}.$$

Here δ_i is the time-series intercept, α_i^{CS} is the cross-sectional pricing error, and λ_{FFR} is the federal-funds innovation risk price. The no-intercept restriction is part of the article's benchmark economic test, not a default regression convention.

The economic mechanism is an ICAPM-style hedging argument. A rate innovation can matter if it captures news about future investment opportunities and if anomaly portfolios differ in their exposure to that news. That statement is weaker than a monetary-policy interpretation. An autoregressive residual may combine policy news, information about the macroeconomy, data revisions, and persistence in the short-rate series. The paper therefore treats the baseline factor as a rate innovation until a separate high-frequency design justifies narrower language.

The article's primary test assets are value-weighted decile portfolios sorted on book-to-market, equity duration, earnings-to-price, long-term reversal, investment-to-assets, property-plant-and-equipment investment, and inventory growth. It also studies the joint system of all seven anomaly families and uses equal-weighted portfolios and additional appendix designs as robustness checks. The benchmark inference reports Shanken-corrected risk-price statistics, bootstrap p-values, a joint pricing-error statistic, and cross-sectional fit metrics.

The mechanism is not one channel. Short-rate innovations may proxy for discount-rate news, cash-flow news, financing-cost changes, working-capital conditions, monetary-policy information, or intertemporal hedging demand. The equity-duration channel is the clearest family-level mechanism because a rate innovation changes the discounting of cash flows at different horizons. For the other anomaly families, the paper predeclares exposure comparisons but does not force a directional sign unless the article-level result and generated portfolio definition support it.

The meaning of a numerical rate innovation can also change across regimes. Conventional positive-rate policy, lower-bound periods, asset-purchase and forward-guidance episodes, and inflation-driven tightening can attach different information to the same measured short-rate movement. The baseline rate residual therefore remains an aggregate state-variable innovation; policy and information language is reserved for the separate high-frequency design.

Some implementation details remain ambiguous at the source-file level. The article identifies the St. Louis Federal Reserve Bank, Kenneth French's data library, Robert Stambaugh's web page, and Lu Zhang as sources, but the exact download files and archive vintages still have to be frozen before the repository can claim exact computational replication.

4 Data and replication design

The data design separates exact inputs, approximate reconstructions, and extension inputs. Exact inputs require a verified article definition, source, vintage, transformation, and estimator role.

Table 1: Mechanisms, hypotheses, tests, and interpretation

Hypothesis	Mechanism	Primary test	Interpretation rule
H1	ICAPM hedging demand	Pricing-error reduction relative to the CAPM	Replication only
H2	Temporal persistence of rate risk	Post-publication compatibility	Sign and magnitude matter
H3	Regime-dependent rate meaning	Interaction equivalence intervals	Non-rejection is insufficient
H4	Aggregate versus decomposed shocks	Component spanning and pricing content	Identification language only
H5	Factor-strength condition	Beta dispersion and rank diagnostics	Weak factors limit interpretation

Source: `research/hypothesis_registry.csv` and `research/mechanism_hypothesis_map.md`.

Approximate reconstructions are documented public or author-data substitutes and are never labelled as exact replication. Extension inputs support post-publication, regime, and high-frequency analyses after their source terms and transformations are frozen.

The baseline sample runs from 1972-01 through 2013-12. The temporal extension begins in 2014-01, with revised-history checks kept separate from the baseline verdict. Every source must record access status, redistribution status, and its role in strict replication before it can support a claim. Exact unavailable inputs lead to a missing-input classification rather than an empirical contradiction.

Table 2: Data sources and sample-construction status

Variable or family	Source	Frequency	Transformation	Status
Article and supplement	Publisher files	Document	Target extraction	Exact local access
Federal funds rate	St. Louis Fed source	Monthly	First-order innovation	Source located, series pending
Treasury-bill rate	St. Louis Fed or author source	Monthly	First-order innovation	Series ambiguous
Market and risk-free returns	Kenneth French library	Monthly	Excess-return factors	Source located, vintage pending
Anomaly decile portfolios	Lu Zhang and cited sources	Monthly	Excess returns after alignment	Files pending
Comparator factors	French, Stambaugh, and q-factor sources	Monthly	Common-intersection factors	Source located, vintage pending
Announcement components	Documented research dataset	Event	Monthly aggregation	Extension input pending

Source: `research/data_access_matrix.csv` and `research/data_dictionary.md`.

Short-rate series are stored in their source units and transformed only in named innovation steps. Market and portfolio returns are aligned to the canonical monthly panel before excess returns, factor construction, or model comparison. Missing returns and shock observations are not forward filled. Comparator models use identical asset-date intersections within each comparison, and delisting or survivorship treatment is recorded from verified source documentation rather than inferred from the data file alone.

Source documents and source registries feed rate series, factor returns, and anomaly portfolios. Those inputs form aligned monthly panels, which then feed baseline replication, diagnostics, temporal extensions, regime tests, and high-frequency decomposition outputs.

Figure 1: Data flow from sources to manuscript evidence
Source: `research/data_flow_diagram.md`.

The replication target is table-level. Each published table or appendix table is mapped to a source location, model, estimator, tolerance rule, and status. A generated statistic can be labelled reproduced only after the same definition and estimator are linked to an auditable output. Nearby public data may be useful for documented reconstruction, but they cannot change the strict replication label.

5 Econometric methods

The baseline empirical design starts with the article's first-order short-rate innovation equations. Alternative autoregressive orders, alternative rate definitions, local-level filters, and announcement-surprise measures are robustness or extension specifications, not replacements for the strict baseline.

First-pass time-series regressions estimate portfolio exposures to the market excess return and the selected short-rate innovation. The notation separates the time-series intercept δ_i, factor loadings β_i, residuals $\varepsilon_{i,t}$, cross-sectional pricing errors α_i^{CS}, and risk prices λ. The second pass is the article's verified no-intercept OLS cross-sectional regression unless a supplement target explicitly calls for an unrestricted zero-beta-rate variant. Estimated-beta uncertainty is attached to the table cell through the Shanken correction or through the separately labelled bootstrap and robustness procedures.

Model comparison reports pricing-error loss and fit on common asset-date intersections. The registered outcomes are root mean squared pricing error, mean absolute pricing error, maximum absolute pricing error, cross-sectional fit, and joint pricing-error tests when the estimator supports them. Statistical significance is not enough: every supported result must also clear the registered economic-magnitude criterion.

Weak-factor diagnostics evaluate whether short-rate betas contain enough cross-sectional variation to support a pricing interpretation. The planned diagnostics include beta dispersion, matrix rank, singular values, condition measures, factor-spanning checks, influence diagnostics, and leave-one-family systems. A short-rate risk price that fails these diagnostics is interpreted as weakly identified even when fit metrics appear favorable.

Temporal and regime methods are designed as extensions. The post-publication analysis has frozen-parameter and refitted variants, with no-lookahead rules for factor construction, portfolio membership, model vintage, and sample intersections. Regime tests use pooled interactions with a declared omitted category, joint tests, equivalence intervals against economic bounds, boundary sensitivity checks, and exploratory unknown-break diagnostics. Stability is not inferred from a failure to reject equality.

The high-frequency decomposition starts from event-level announcement observations only after source terms and component definitions are frozen. The design records the event window, policy, information, and ambiguous components, the monthly aggregation rule, no-event month handling, spanning tests, and incremental pricing tests. Causal language is confined to this identified event design.

Out-of-sample evaluation defines the forecast origin, training window, refit schedule, benchmark models, loss functions, and interpretation before results are inspected. Multiplicity adjustments are predeclared within baseline pricing, temporal extension, regime stability, shock decomposition, and weak-factor families.

Table 3: Method-to-hypothesis map

Hypothesis	Primary method	Decision rule
H1	No-intercept two-pass pricing on the article intersection	Pricing-error improvement must be economically material
H2	Frozen and refitted post-publication evaluation	Sign, magnitude, and loss compatibility must hold
H3	Pooled regime interactions with equivalence intervals	Estimates must remain inside economic bounds
H4	Component spanning and pricing tests	Components must not add pricing content for sufficiency
H5	Beta dispersion, rank, and robust-inference diagnostics	Strength gates must pass before interpretation

Source: `research/statistical_protocol.md` and `research/hypothesis_registry.csv`.

6 Baseline replication results

Baseline empirical results are pending. The current table-level audit classifies the frozen targets as not reproducible because required generated baseline artifacts are missing. For this status statement, the sample is the strict monthly baseline, the model set is the registered baseline and comparator system, the estimator is the table-level audit over generated cells, the test assets are the registered anomaly portfolios, uncertainty is unavailable because generated standard errors do not exist, and economic magnitude is unavailable because no replicated table cell has been produced.

This pending status does not evaluate the published evidence. It states only that the repository cannot yet generate the necessary baseline tables from source-compatible inputs. Once the missing inputs are registered, the same audit will classify each target as reproduced, approximately reproduced, contradicted, not attempted, or still blocked by missing input.

7 Robustness and weak-factor diagnostics

The robustness section will report whether any baseline pricing result survives alternative rate innovations, covariance estimators, portfolio weighting, influence checks, and weak-factor diagnostics. The first question is mechanical: do estimated short-rate betas vary enough across assets for the second pass to identify a risk price? The second question is economic: does the factor reduce pricing errors by a material amount relative to the comparator models?

Robustness results are pending because baseline estimates are pending. For this status statement, the sample is the generated strict baseline, the model is the registered robustness family, the estimator is diagnostic comparison over baseline pricing artifacts, the test assets are the replication-audit assets, uncertainty is governed by weak-factor and familywise diagnostics, and economic magnitude is blocked until baseline estimates exist.

8 Temporal and monetary-regime evidence

The temporal extension asks whether the baseline relation is stable after the publication sample. The baseline endpoint remains 2013-12, and the extension starts in 2014-01. The design prevents revised data or later observations from changing the baseline replication label.

The regime extension asks whether short-rate betas or risk prices differ across predeclared monetary regimes. The stability claim is deliberately demanding: ordinary non-rejection of equality is not sufficient. The paper will report interaction estimates, break diagnostics, confidence intervals, and predeclared stability classifications once extension-compatible panels exist.

Temporal and regime results are pending. For this status statement, the sample is the extension-compatible monthly panel, the model is the temporal and regime-interaction system, the estimator is predeclared temporal comparison plus regime break diagnostics, the test assets are source-compatible anomaly portfolios, uncertainty uses the registered temporal, Wald, Holm-adjusted, and break-test procedures, and economic magnitude is blocked until extension inputs exist.

9 Policy and information shock decomposition

The high-frequency extension is the only section in which policy-shock language is allowed. Its purpose is to test whether the aggregate rate innovation used in the baseline masks distinct policy, central-bank-information, and ambiguous components. The planned design uses event-level observations, preserves the component classification, aggregates to monthly frequency under a predeclared rule, and compares the decomposed shocks with the baseline rate innovation.

This section is not yet an identification result. The legally usable event-level data and frozen decomposition artifacts are not available, so the paper cannot state whether policy or information components price anomaly returns. For this status statement, the sample is the event-level announcement panel aggregated to months after source terms are registered, the model is the component decomposition, the estimator is monthly aggregation plus spanning and pricing checks, the test assets are compatible anomaly portfolios, uncertainty follows the registered shock-decomposition diagnostics, and economic magnitude is blocked until event data are registered.

10 Discussion and limitations

The main limitation is input availability. Exact rate definitions, portfolio files, generated factor panels, baseline estimator outputs, extension panels, and event-level shock data are not yet complete. This limitation is a research status condition, not evidence for or against the baseline article.

The second limitation is interpretive. Even with a replicated risk price, a priced covariance with a rate innovation would not by itself establish a narrow monetary-policy mechanism. The design therefore separates pricing evidence from identification evidence and treats high-frequency decomposition as an extension.

The third limitation is econometric. Cross-sectional pricing is fragile when betas have little dispersion, samples are short within regimes, or comparator models are evaluated on different asset-date intersections. The paper's diagnostic structure is intended to make those limits visible before substantive interpretation.

11 Conclusion

This preliminary draft converts the repository protocol into an academic paper structure while preserving the evidence discipline required for replication. The paper will first ask whether the published short-rate innovation pricing evidence can be reproduced from source-compatible inputs. It will then ask whether the relation is stable over time, across monetary regimes, and after decomposing aggregate rate innovations into high-frequency components. No empirical conclusion is reported until the registered inputs and generated artifacts are available.

A Table-level replication audit

The table-level audit maps the article and supplement targets to replication statuses. The current audit covers the article tables and supplement appendix tables through Table A.14, and every target is blocked by missing generated baseline inputs. The generated audit records the current missing-input classification for each target.

Table 4: Original table groups and replication targets

Original evidence group	Registered targets	Current classification
Article descriptive and pricing tables	TBL_001–TBL_009	Missing generated inputs
Supplement interest-rate robustness	APP_TBL_A01–APP_TBL_A06	Missing generated inputs
Supplement portfolio and model extensions	APP_TBL_A07–APP_TBL_A14	Missing generated inputs

Source: `research/table_target_manifest.csv`.

B Data access and reproducibility

Restricted article and supplement files remain local-only. The committed repository records metadata and source-access status, but it does not redistribute copyrighted source documents, prompt files, licensed data extracts, local catalogs, temporary artifacts, or machine-specific environment manifests. The source-only release gate reports zero critical restricted-path issues and one major missing-input issue.

Source-access requirements, release artifacts, and validation status are recorded outside the main argument in the repository documentation and generated release manifests.

C Hypothesis registry

The confirmatory registry contains hypotheses for baseline replication, post-publication compatibility, regime equivalence, shock decomposition, and weak-factor strength. The registry distinguishes null, alternative, primary outcome, estimator, sample, test assets, economic direction or equivalence criterion, multiplicity family, and supported, unsupported, or inconclusive interpretation. Exploratory structural-break alignment is recorded separately from confirmatory claims.

D Table and figure architecture

The manuscript table and figure map defines the planned main-text evidence sequence before result writing begins. The architecture prioritizes a source-status table, baseline diagnostic table, replication-comparison table, weak-factor diagnostic table, baseline pricing table, comparator table, temporal-extension table, regime-stability table, and shock-decomposition table. The planned figures cover rate innovations and regimes, beta-return relations, rolling or expanding risk prices, pricing-error comparisons, regime equivalence intervals, and shock-spanning results.

The LaTeX shells and generation contracts contain no empirical cells and cannot be mistaken for output. Every shell is linked to a machine-readable source and an evidence gate. Result writing remains blocked until the required generated artifact exists.

E Detailed out-of-sample design

The out-of-sample design freezes an initial training endpoint at 1999-12 and uses a predeclared annual refit schedule. Forecast records must store model vintage, training dates, factor definition, asset universe, benchmark model, and loss function. Results from this design will be interpreted as evidence about stability of pricing errors, not as a substitute for the baseline replication audit.

F Additional diagnostics

Additional diagnostics include beta dispersion, singular values, rank checks, factor spanning, influence diagnostics, leave-one-family systems, boundary-shift checks, and multiple-testing adjustments. Diagnostic outputs remain pending until the relevant baseline and extension artifacts exist.

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